COMP5400: Bio-Inspired Computation
Coursework 1:
Evolution and Co-evolution with BEAST
Due Date: 3 March, 2026

Module Leader: Netta Cohen

Marking

This coursework is summative and is worth 40% of the total grade for this module, and the time allotted to it should be commensurate.

Courseware

This coursework is based on "demos" in the module courseware: BEAST. You may use either the C++ version (Beast) or the Python version (PyBEAST). Un update to PyBEAST (pyBEAST++) is also available on Minerva. Software will be supported when used on School of Computer Science machines.

Assessment

Please submit your answers to questions as a single pdf on on Turnitin. In addition, please submit your modified code to Gradescope. You should include:

- A pdf of typed answers to questions. Hand-written answers will not be accepted. Please use arial fonts with size 11pt. The PDF should mention which version of BEAST you are using. In your answers, please include all evidence, e.g. graphics, or short extracts of code (please avoid screen shots or code printed on a dark background). Figures must be of sufficient size and clearly labeled (including e.g., axis label and ticks, in a legible font, min. size 9pt).
- The specific code modifications (not the entire file) that you made to answer Q4 should also be included in your answer sheet to Q4. To include these modifications: please cut and paste the relevant lines of code into your answer sheet, rather than using screen shots. Please avoid screenshots with a black background or with any text colour that is not clearly legible in the pdf of your answer sheet.
- Keep your answers to questions concise, clear, evidenced and well-reasoned.
- The modified demo files for Q4 should be submitted to Gradescope. The code will be checked that it compiles and runs properly.

Questions

Part 0 – Formative: This part of the assignment is optional, but encouraged. You are encouraged to work with your classmates. The purpose of this component is to acquaint you with BEAST, and to introduce you to Braitenberg vehicles (which we will study in detail in later lectures).

1. Run the Braitenberg demo in Beast. For these questions (in Part 0) you need not look at the code, or look any material up. Just watch the demo: you will see agents roaming around the arena. What are they doing? Do they have "purpose"? If so, try to articulate it. Do you think these agents have an artificial brain? What evidence do you have for your conclusion (either way).

The remainder of the worksheet is summative.

Part I: Run the **mouse** demo in BEAST.

1. Plot the fitness of the mice as a function of the generation count. Be judicious where you terminate your simulation. Don't stop too early. Briefly motivate why you show the particular range you chose.

[5 marks]

2. Describe the **behaviour** of the mice as it evolves over evolutionary time.

[5 marks]

3. Try to improve the performance of the mice by experimenting with <u>physical</u> components or configurations of the agents, with focus on:

a) sensor type or sensor configurations (e.g., experiment with sensor parameters, create your own sensors, or combine existing sensors). If the sensors require it, you may need to consider the interface to the neural network of the "mouse", [15 marks]

b) the neural architecture. [15 marks]

<u>For both a) and b)</u>: Experiments should be well motivated and systematically evaluated. Explain the reasoning behind your experiments, motivate your design, quantify performance and draw conclusions. Iterations of experiments may be useful. If the changes you introduce have both advantages and disadvantages, consider the resulting trade-offs. Submit your scripts, including seeds and parameter values, and output for the final generation simulated.

<u>Note 1</u>: changing the maximum speed of the mice will not count as an improvement.
<u>Note 2</u>: clearly distinguish between your experiments in parts a) and b).

4. Describe the fitness function used by the mice in the evolutionary algorithm (look in the code to inform your answer). Experiment with the fitness function: change it and run the simulation again. You are encouraged to be creative, so that the fitness function assesses very different aspects of the agent performance or behaviour. Judging from the behaviour of the mice, explain whether you believe that the performance of the mice improves or deteriorates with your changes.

[10 marks]

5. How would you define the performance of mice <u>objectively</u> when using different fitness functions? (In other words, how would you compare the relative performance of the mice given different fitness functions?) Implement and evaluate your proposed objective measure.

[5 marks]

6. Based on the lecture notes and the literature cited there, discuss what you understand by the term “collective behaviour”. Describe and explain the behaviour of the mice. Do you see evidence of collective behaviour? Explain, argue and evidence your answer.

[15 marks]

Part II: Run the **chase** demo in BEAST. Run three demonstrations for 100-200 generations, each. (You can do this in parallel.) Note: make sure the seed is not fixed between runs. Record the behaviour of both types of agent after 10, 50 and 100 generations (and the maximum number of generations, if longer). Let one simulation run for longer (until you have evidence that the performance has peaked). You may share data of long simulations with classmates to view results using different seeds.

7. Plot the fitness of each agent as a function of the generation count for both types of agent for all runs. Think of a good way to represent, visualise and compare your results in plots. Include results from long simulations received from your classmates. To give proper credit to others, state the full names of students who have shared data with you.

[10 marks]

8. Describe the behaviour of the agents. Is this an example of co-evolution? Argue. You should explain your reasons and evidence your answer (for co-evolution or for its absence) based on the behaviour of the agents after different numbers of generations.

[5 marks]

9. Judging the fitness plots for the same simulations, do you see evidence for co-evolution? Discuss and evidence your answers.

[5 marks]

10. Is the behaviour of the agent “intelligent”? Argue and reason your answer, and give your own motivation and evidence (for intelligence or for its absence, or for inconclusive evidence). In your answer, consider definitions of intelligence and artificial intelligence. Some definitions may come from the scientific literature, for example the paper ‘Intelligence without Representation’ by Rodney Brooks, which can be found on Minerva. If you wish (note, this is not required) you may argue to reject the definitions proposed by others and propose your own definition of intelligence.

[10 marks]

[100 marks total]